Appendix B
Answers to Odd-Numbered Non-Review Exercises

Chapter 1

1.1 (a) Sample size = 15

(b) Sample mean = 3.787

(c) Sample median = 3.6

(e) $\bar{x}_{\text{tr}(20)} = 3.678$

(f) They are about the same.

1.3 (b) Yes, the aging process has reduced the tensile strength.

(c) $\bar{x}_{\text{Aging}} = 209.90$, $\bar{x}_{\text{No aging}} = 222.10$

(d) $\tilde{x}_{\text{Aging}} = 210.00$, $\tilde{x}_{\text{No aging}} = 221.50$. The means and medians are similar for each group.

1.5 (b) Control: $\bar{x} = 5.60$, $\tilde{x} = 5.00$, $\bar{x}_{\text{tr}(10)} = 5.13$. Treatment: $\bar{x} = 7.60$, $\tilde{x} = 4.50$, $\bar{x}_{\text{tr}(10)} = 5.63$.

(c) The extreme value of 37 in the treatment group plays a strong leverage role for the mean calculation.

1.7 Sample variance = 0.943
Sample standard deviation = 0.971

1.9 (a) No aging: sample variance = 23.66, sample standard deviation = 4.86. Aging: sample variance = 42.10, sample standard deviation = 6.49.

(b) Based on the numbers in (a), the variation in "Aging" is smaller than the variation in "No aging," although the difference is not so apparent in the plot.

1.11 Control: sample variance = 69.38, sample standard deviation = 8.33. Treatment: sample variance = 128.04, sample standard deviation = 11.32.

1.13 (a) Mean = 124.3, median = 120

(b) 175 is an extreme observation.

1.15 Yes, P-value = 0.03125, probability of obtaining $HHHHH$ with a fair coin.

1.17 (a) The sample means for nonsmokers and smokers are 30.32 and 43.70, respectively.

(b) The sample standard deviations for nonsmokers and smokers are 7.13 and 16.93, respectively.

(d) Smokers appear to take a longer time to fall asleep. For smokers the time to fall asleep is more variable.

1.19 (a)

Stem	Leaf	Frequency
0	22233457	8
1	023558	6
2	035	3
3	03	2
4	057	3
5	0569	4
6	0005	4

(b)

Class Interval	Class Midpoint	Freq.	Rel. Freq.
0.0–0.9	0.45	8	0.267
1.0–1.9	1.45	6	0.200
2.0–2.9	2.45	3	0.100
3.0–3.9	3.45	2	0.067
4.0–4.9	4.45	3	0.100
5.0–5.9	5.45	4	0.133
6.0–6.9	6.45	4	0.133

(c) Sample mean = 2.7967
Sample range = 6.3
Sample standard deviation = 2.2273

1.21 (a) $\bar{x} = 74.02$ and $\tilde{x} = 78$

(b) $s = 39.26$

1.23 (b) $\bar{x}_{1980} = 395.10$, $\bar{x}_{1990} = 160.15$

(c) The mean emissions dropped between 1980 and 1990; the variability also decreased because there were no longer extremely large emissions.

1.25 (a) Sample mean = 33.31

(b) Sample median = 26.35

(d) $\bar{x}_{\text{tr}(10)} = 30.97$

Chapter 2

2.1 (a) $S = \{8, 16, 24, 32, 40, 48\}$

(b) $S = \{-5, 1\}$

(c) $S = \{T, HT, HHT, HHH\}$

(d) S ={Africa, Antarctica, Asia, Australia, Europe, North America, South America}

(e) $S = \phi$

2.3 $A = C$

2.5 Using the tree diagram, we obtain

$S = \{1HH,\ 1HT,\ 1TH,\ 1TT,\ 2H,\ 2T,\ 3HH,\ 3HT,\ 3TH,\ 3TT,\ 4H,\ 4T,\ 5HH,\ 5HT,\ 5TH,\ 5TT,\ 6H,\ 6T\}$

2.7 $S_1 = \{MMMM, MMMF, MMFM, MFMM, FMMM, MMFF, MFMF, MFFM, FMFM, FFMM, FMMF, MFFF, FMFF, FFMF, FFFM, FFFF\}$;
$S_2 = \{0, 1, 2, 3, 4\}$

2.9 (a) $A = \{1HH, 1HT, 1TH, 1TT, 2H, 2T\}$

(b) $B = \{1TT, 3TT, 5TT\}$

(c) $A^{'} = \{3HH, 3HT, 3TH, 3TT, 4H, 4T, 5HH, 5HT, 5TH, 5TT, 6H, 6T\}$

(d) $A^{'} \cap B = \{3TT, 5TT\}$

(e) $A \cup B = \{1HH, 1HT, 1TH, 1TT, 2H, 2T, 3TT, 5TT\}$

2.11 (a) $S = \{M_1M_2, M_1F_1, M_1F_2, M_2M_1, M_2F_1, M_2F_2, F_1M_1, F_1M_2, F_1F_2, F_2M_1, F_2M_2, F_2F_1\}$

(b) $A = \{M_1M_2, M_1F_1, M_1F_2, M_2M_1, M_2F_1, M_2F_2\}$

(c) $B = \{M_1F_1, M_1F_2, M_2F_1, M_2F_2, F_1M_1, F_1M_2, F_2M_1, F_2M_2\}$

(d) $C = \{F_1F_2, F_2F_1\}$

(e) $A \cap B = \{M_1F_1, M_1F_2, M_2F_1, M_2F_2\}$

(f) $A \cup C = \{M_1M_2, M_1F_1, M_1F_2, M_2M_1, M_2F_1, M_2F_2, F_1F_2, F_2F_1\}$

2.15 (a) {nitrogen, potassium, uranium, oxygen}

(b) {copper, sodium, zinc, oxygen}

(c) {copper, sodium, nitrogen, potassium, uranium, zinc}

(d) {copper, uranium, zinc}

(e) ϕ

(f) {oxygen}

2.19 (a) The family will experience mechanical problems but will receive no ticket for a traffic violation and will not arrive at a campsite that has no vacancies.

(b) The family will receive a traffic ticket and arrive at a campsite that has no vacancies but will not experience mechanical problems.

(c) The family will experience mechanical problems and will arrive at a campsite that has no vacancies.

(d) The family will receive a traffic ticket but will not arrive at a campsite that has no vacancies.

(e) The family will not experience mechanical problems.

2.21 18

2.23 156

2.25 20

2.27 48

2.29 210

2.31 72

2.33 (a) 1024; (b) 243

2.35 362,880

2.37 2880

2.39 (a) 40,320; (b) 336

2.41 360

2.43 24

2.45 3360

2.47 56

2.49 (a) Sum of the probabilities exceeds 1.
(b) Sum of the probabilities is less than 1.
(c) A negative probability
(d) Probability of both a heart and a black card is zero.

2.51 $S = \{\$10, \$25, \$100\}$; $P(10) = \frac{11}{20}$, $P(25) = \frac{3}{10}$, $P(100) = \frac{15}{100}$; $\frac{17}{20}$

2.53 (a) 0.3; (b) 0.2

2.55 10/117

2.57 (a) 5/26; (b) 9/26; (c) 19/26

2.59 (a) 94/54,145; (b) 143/39,984

2.61 (a) 22/25; (b) 3/25; (c) 17/50

2.63 (a) 0.32; (b) 0.68; (c) office or den

2.65 (a) 0.8; (b) 0.45; (c) 0.55

2.67 (a) 0.31; (b) 0.93; (c) 0.31

2.69 (a) 0.009; (b) 0.999; (c) 0.01

2.71 (a) 0.048; (b) \$50,000; (c) \$12,500

2.73 (a) The probability that a convict who pushed dope also committed armed robbery.
(b) The probability that a convict who committed armed robbery did not push dope.
(c) The probability that a convict who did not push dope also did not commit armed robbery.

2.75 (a) 14/39; (b) 95/112

2.77 (a) 5/34; (b) 3/8

2.79 (a) 0.018; (b) 0.614; (c) 0.166; (d) 0.479

2.81 (a) 0.35; (b) 0.875; (c) 0.55

2.83 (a) 9/28; (b) 3/4; (c) 0.91

2.85 0.27

2.87 5/8

2.89 (a) 0.0016; (b) 0.9984

2.91 (a) 91/323; (b) 91/323

2.93 (a) 0.75112; (b) 0.2045

2.95 0.0960

2.97 0.40625

2.99 0.1124

2.101 0.857

Chapter 3

3.1 Discrete; continuous; continuous; discrete; discrete; continuous

3.3

Sample Space	w
HHH	3
HHT	1
HTH	1
THH	1
HTT	-1
THT	-1
TTH	-1
TTT	-3

3.5 (a) 1/30; (b) 1/10

3.7 (a) 0.68; (b) 0.375

3.9 (b) 19/80

3.11

x	0	1	2
$f(x)$	$\frac{2}{7}$	$\frac{4}{7}$	$\frac{1}{7}$

3.13
$$F(x) = \begin{cases} 0, & \text{for } x < 0, \\ 0.41, & \text{for } 0 \le x < 1, \\ 0.78, & \text{for } 1 \le x < 2, \\ 0.94, & \text{for } 2 \le x < 3, \\ 0.99, & \text{for } 3 \le x < 4, \\ 1, & \text{for } x \ge 4 \end{cases}$$

3.15
$$F(x) = \begin{cases} 0, & \text{for } x < 0, \\ \frac{2}{7}, & \text{for } 0 \le x < 1, \\ \frac{6}{7}, & \text{for } 1 \le x < 2, \\ 1, & \text{for } x \ge 2 \end{cases}$$
(a) 4/7; (b) 5/7

3.17 (b) 1/4; (c) 0.3

3.19 $F(x) = \begin{cases} 0, & x < 1 \\ \frac{x-1}{2}, & 1 \le x < 3 \\ 1, & x \ge 3 \end{cases}$; 1/4

3.21 (a) 3/2; (b) $F(x) = \begin{cases} 0, & x < 0 \\ x^{3/2}, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$; 0.3004

3.23
$$F(w) = \begin{cases} 0, & \text{for } w < -3, \\ \frac{1}{27}, & \text{for } -3 \le w < -1, \\ \frac{7}{27}, & \text{for } -1 \le w < 1, \\ \frac{19}{27}, & \text{for } 1 \le w < 3, \\ 1, & \text{for } w \ge 3 \end{cases}$$
(a) 20/27; (b) 2/3

3.25

t	20	25	30
$P(T = t)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{5}$

3.27 (a)
$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - \exp(-x/2000), & x \ge 0 \end{cases}$$
(b) 0.6065; (c) 0.6321

3.29 (b)
$$F(x) = \begin{cases} 0, & x < 1, \\ 1 - x^{-3}, & x \ge 1 \end{cases}$$
(c) 0.0156

3.31 (a) 0.2231; (b) 0.2212

3.33 (a) $k = 280$; (b) 0.3633; (c) 0.0563

3.35 (a) 0.1528; (b) 0.0446

3.37 (a) 1/36; (b) 1/15

3.39 (a)

	$f(x,y)$	x = 0	1	2	3
y	0	0	$\frac{3}{70}$	$\frac{9}{70}$	$\frac{3}{70}$
	1	$\frac{2}{70}$	$\frac{18}{70}$	$\frac{18}{70}$	$\frac{2}{70}$
	2	$\frac{3}{70}$	$\frac{9}{70}$	$\frac{3}{70}$	0

(b) 1/2

3.41 (a) 1/16; (b) $g(x) = 12x(1-x)^2$, for $0 \le x \le 1$; (c) 1/4

3.43 (a) 3/64; (b) 1/2

3.45 0.6534

3.47 (a) Dependent; (b) 1/3

3.49 (a)

x	1	2	3
$g(x)$	0.10	0.35	0.55

(b)

y	1	2	3
$h(y)$	0.20	0.50	0.30

(c) 0.2857

3.51 (a)

	$f(x,y)$	x = 0	1	2	3
y	0	$\frac{1}{55}$	$\frac{6}{55}$	$\frac{6}{55}$	$\frac{1}{55}$
	1	$\frac{6}{55}$	$\frac{16}{55}$	$\frac{6}{55}$	0
	2	$\frac{6}{55}$	$\frac{6}{55}$	0	0
	3	$\frac{1}{55}$	0	0	0

(b) 42/55

3.53 5/8

3.55 Independent

3.57 (a) 3; (b) 21/512

3.59 Dependent

Chapter 4

4.1 0.88

4.3 25¢

4.5 \$1.23

4.7 \$500

4.9 \$6900

4.11 $(\ln 4)/\pi$

4.13 100 hours

4.15 0

4.17 209

4.19 \$1855

4.21 \$833.33

4.23 (a) 35.2; (b) $\mu_X = 3.20$, $\mu_Y = 3.00$

4.25 2

4.27 2000 hours

4.29 (b) 3/2

4.31 (a) 1/6; (b) $(5/6)^5$

4.33 \$5,250,000

4.35 0.74

4.37 1/18; in terms of actual profit, the variance is $\frac{1}{18}(5000)^2$

4.39 1/6

4.41 118.9

4.43 $\mu_Y = 10$; $\sigma_Y^2 = 144$

4.45 0.01

4.47 -0.0062

4.49 $\sigma_X^2 = 0.8456$, $\sigma_X = 0.9196$

4.51 $-1/\sqrt{5}$

4.53 $\mu_{g(X)} = 10.33$, $\sigma_{g(X)} = 6.66$

4.55 \$0.80

4.57 209

4.59 $\mu = 7/2$, $\sigma^2 = 15/4$

4.61 3/14

4.63 52

4.65 (a) 7; (b) 0; (c) 12.25

4.67 46/63

4.69 (a) $E(X) = E(Y) = 1/3$ and $\text{Var}(X) = \text{Var}(Y) = 4/9$; (b) $E(Z) = 2/3$ and $\text{Var}(Z) = 8/9$

4.71 (a) 4; (b) 32; 16

4.73 By direct calculation, $E(e^Y) = 1884.32$. Using the second-order approximation, $E(e^Y) \approx 1883.38$, which is very close to the true value.

4.75 0.03125

4.77 (a) At most 4/9; (b) at least 5/9; (c) at least 21/25; (d) 10

Chapter 5

5.1 $\mu = \frac{1}{k}\sum_{i=1}^{k} x_i$, $\sigma^2 = \frac{1}{k}\sum_{i=1}^{k}(x_i - \mu)^2$

5.3 $f(x) = \frac{1}{10}$, for $x = 1, 2, \ldots, 10$, and $f(x) = 0$ elsewhere; 3/10

5.5 (a) 0.0480; (b) 0.2375; (c) $P(X = 5 \mid p = 0.3) = 0.1789$, $P = 0.3$ is reasonable.

5.7 (a) 0.0474; (b) 0.0171

5.9 (a) 0.7073; (b) 0.4613; (c) 0.1484

5.11 0.1240

5.13 0.8369

5.15 (a) 0.0778; (b) 0.3370; (c) 0.0870

5.17 $\mu = 3.5$, $\sigma^2 = 1.05$

5.19 $f(x_1, x_2, x_3) = \binom{n}{x_1, x_2, x_3} 0.35^{x_1} 0.05^{x_2} 0.60^{x_3}$

5.21 0.0095

5.23 0.0077

5.25 0.8670

5.27 (a) 0.2852; (b) 0.9887; (c) 0.6083

5.29 5/14

5.31 $h(x; 6, 3, 4) = \dfrac{\binom{4}{x}\binom{2}{3-x}}{\binom{6}{3}}$, for $x = 1, 2, 3$;

$P(2 \le X \le 3) = 4/5$

5.33 (a) 0.3246; (b) 0.4496

5.35 0.9517

5.37 (a) 0.6815; (b) 0.1153

5.39 0.9453

5.41 0.6077

5.43 (a) 4/33; (b) 8/165

5.45 0.2315

5.47 (a) 0.3991; (b) 0.1316

5.49 0.0515

5.51 63/64

5.53 (a) 0.3840; (b) 0.0067

5.55 (a) 0.0630; (b) 0.9730

5.57 (a) 0.1429; (b) 0.1353

5.59 (a) 0.1638; (b) 0.032

5.61 0.2657

5.63 $\mu = 6$, $\sigma^2 = 6$

5.65 (a) 0.2650; (b) 0.9596

5.67 (a) 0.8243; (b) 14

5.69 4

5.71 5.53×10^{-4}; $\mu = 7.5$

5.73 (a) 0.0137; (b) 0.0830

5.75 0.4686

Chapter 6

6.3 (a) 0.6; (b) 0.7; (c) 0.5

6.5 (a) 0.0823; (b) 0.0250; (c) 0.2424; (d) 0.9236; (e) 0.8133; (f) 0.6435

6.7 (a) 0.54; (b) -1.72; (c) 1.28

6.9 (a) 0.1151; (b) 16.1; (c) 20.275; (d) 0.5403

6.11 (a) 0.0548; (b) 0.4514; (c) 23 cups; (d) 189.95 milliliters

6.13 (a) 0.8980; (b) 0.0287; (c) 0.6080

6.15 (a) 0.0571; (b) 99.11%; (c) 0.3974; (d) 27.952 minutes; (e) 0.0092

6.17 6.24 years

6.19 (a) 51%; (b) \$18.37

6.21 (a) 0.0401; (b) 0.0244

6.23 26 students

6.25 (a) 0.3085; (b) 0.0197

6.27 (a) 0.9514; (b) 0.0668

6.29 (a) 0.1171; (b) 0.2049

6.31 0.1357

6.33 (a) 0.0778; (b) 0.0571; (c) 0.6811

6.35 (a) 0.8749; (b) 0.0059

6.37 (a) 0.0228; (b) 0.3974

6.41 $2.8e^{-1.8} - 3.4e^{-2.4} = 0.1545$

6.43 (a) $\mu = 6$; $\sigma^2 = 18$; (b) between 0 and 14.485 million liters

6.45 $\sum_{x=4}^{6} \binom{6}{x} (1 - e^{-3/4})^x (e^{-3/4})^{6-x} = 0.3968$

6.47 (a) $\sqrt{\pi/2} = 1.2533$ years; (b) e^{-2}

6.49 (a) Mean $= 0.25$, median $= 0.206$; (b) variance $= 0.0375$; (c) 0.2963

6.51 $e^{-4} = 0.0183$

6.53 (a) $\mu = \alpha\beta = 50$; (b) $\sigma^2 = \alpha\beta^2 = 500$; $\sigma = \sqrt{500}$; (c) 0.815

6.55 (a) 0.1889; (b) 0.0357

6.57 Mean $= e^6$, variance $= e^{12}(e^4 - 1)$

6.59 (a) e^{-5}; (b) $\beta = 0.2$

Chapter 7

7.1 $g(y) = 1/3$, for $y = 1, 3, 5$

7.3
$$g(y_1, y_2) = \binom{2}{\frac{y_1+y_2}{2}, \frac{y_1-y_2}{2}, 2 - y_1} \times \left(\frac{1}{4}\right)^{(y_1+y_2)/2} \left(\frac{1}{3}\right)^{(y_1-y_2)/2} \left(\frac{5}{12}\right)^{2-y_1};$$
for $y_1 = 0, 1, 2$; $y_2 = -2, -1, 0, 1, 2$; $y_2 \le y_1$; $y_1 + y_2 = 0, 2, 4$

7.7 Gamma distribution with $\alpha = 3/2$ and $\beta = m/2b$

7.9 (a) $g(y) = 32/y^3$, for $y > 4$; (b) 1/4

7.11 $h(z) = 2(1 - z)$, for $0 < z < 1$

7.13 $h(w) = 6 + 6w - 12w^{1/2}$, for $0 < w < 1$

7.15 $g(y) = \begin{cases} \frac{2}{9\sqrt{y}}, & 0 < y < 1, \\ \frac{\sqrt{y}+1}{9\sqrt{y}}, & 1 < y < 4 \end{cases}$

7.19 Both equal μ.

7.23 (a) Gamma$(2, 1)$; (b) Uniform$(0, 1)$

Chapter 8

8.1 (a) Responses of all people in Richmond who have a telephone;
(b) Outcomes for a large or infinite number of tosses of a coin;
(c) Length of life of such tennis shoes when worn on the professional tour;
(d) All possible time intervals for this lawyer to drive from her home to her office.

8.3 (a) $\bar{x} = 3.2$ seconds; (b) $\tilde{x} = 3.1$ seconds

8.5 (a) $\bar{x} = 2.4$; (b) $\tilde{x} = 2$; (c) $m = 3$

8.7 (a) 53.75; (b) 75 and 100

8.9 (a) Range is 10; (b) $s = 3.307$

8.11 (a) 2.971; (b) 2.971

8.13 $s = 0.585$

8.15 (a) 45.9; (b) 5.1

8.17 0.3159

8.19 (a) Variance is reduced from 0.49 to 0.16.
(b) Variance is increased from 0.04 to 0.64.

8.21 Yes

8.23 (a) $\mu = 5.3$; $\sigma^2 = 0.81$;
(b) $\mu_{\bar{X}} = 5.3$; $\sigma^2_{\bar{X}} = 0.0225$;
(c) 0.9082

8.25 (a) 0.6898; (b) 7.35

8.29 0.5596

8.31 (a) The chance that the difference in mean drying time is larger than 1.0 is 0.0013; (b) 13

8.33 (a) 1/2; (b) 0.3085

8.35 $P(\bar{X} \leq 775 \mid \mu = 760) = 0.9332$

8.37 (a) 27.488; (b) 18.475; (c) 36.415

8.39 (a) 0.297; (b) 32.852; (c) 46.928

8.41 (a) 0.05; (b) 0.94

8.45 (a) 0.975; (b) 0.10; (c) 0.875; (d) 0.99

8.47 (a) 2.500; (b) 1.319; (c) 1.714

8.49 No; $\mu > 20$

8.51 (a) 2.71; (b) 3.51; (c) 2.92;
(d) 0.47; (e) 0.34

8.53 The F-ratio is 1.44. The variances are not significantly different.

Chapter 9

9.1 56

9.3 $0.3097 < \mu < 0.3103$

9.5 (a) $22,496 < \mu < 24,504$; (b) error ≤ 1004

9.7 35

9.9 $10.15 < \mu < 12.45$

9.11 $0.978 < \mu < 1.033$

9.13 $47.722 < \mu < 49.278$

9.15 $(13,075, 33,925)$

9.17 $(6.05, 16.55)$

9.19 323.946 to 326.154

9.21 Upper prediction limit: 9.42;
upper tolerance limit: 11.72

9.25 Yes, the value of 6.9 is outside of the prediction interval.

9.27 (a) $(0.9876, 1.0174)$;
(b) $(0.9411, 1.0639)$;
(c) $(0.9334, 1.0716)$

9.35 $2.9 < \mu_1 - \mu_2 < 7.1$

9.37 $2.80 < \mu_1 - \mu_2 < 3.40$

9.39 $1.5 < \mu_1 - \mu_2 < 12.5$

9.41 $0.70 < \mu_1 - \mu_2 < 3.30$

9.43 $-6536 < \mu_1 - \mu_2 < 2936$

9.45 $(-0.74, 6.30)$

9.47 $(-6.92, 36.70)$

9.49 $0.54652 < \mu_B - \mu_A < 1.69348$

9.51 Method 1: $0.194 < p < 0.262$; method 2: $0.1957 < p < 0.2639$

9.53 (a) $0.498 < p < 0.642$; (b) error ≤ 0.072

9.55 (a) $0.739 < p < 0.961$; (b) no

9.57 (a) $0.644 < p < 0.690$; (b) error ≤ 0.023

9.59 2576

9.61 160

9.63 9604

9.65 $-0.0136 < p_F - p_M < 0.0636$

9.67 $0.0011 < p_1 - p_2 < 0.0869$

9.69 $(-0.0849, 0.0013)$; not significantly different

9.71 $0.293 < \sigma^2 < 6.736$; valid claim

9.73 $3.472 < \sigma^2 < 12.804$

9.75 $9.27 < \sigma < 34.16$

9.77 $0.549 < \sigma_1/\sigma_2 < 2.690$

9.79 $0.016 < \sigma_1^2/\sigma_2^2 < 0.454$; no

9.81 $\frac{1}{n}\sum_{i=1}^{n} x_i$

9.83 $\hat{\beta} = \bar{x}/5$

9.85 $\hat{\theta} = \max\{x_1, \ldots, x_n\}$

9.87 $x \ln p + (1-x)\ln(1-p)$. Set the derivative with respect to $p = 0$; $\hat{p} = x = 1.0$

Chapter 10

10.1 (a) Conclude that less than 30% of the public is allergic to some cheese products when, in fact, 30% or more is allergic.
(b) Conclude that at least 30% of the public is allergic to some cheese products when, in fact, less than 30% is allergic.

10.3 (a) The firm is not guilty;
(b) the firm is guilty.

10.5 (a) 0.0559;
(b) $\beta = 0.0017$; $\beta = 0.00968$; $\beta = 0.5557$

10.7 (a) 0.1286;
(b) $\beta = 0.0901$; $\beta = 0.0708$.
(c) The probability of a type I error is somewhat large.

10.9 (a) $\alpha = 0.0850$; (b) $\beta = 0.3410$

10.11 (a) $\alpha = 0.1357$; (b) $\beta = 0.2578$

10.13 $\alpha = 0.0094$; $\beta = 0.0122$

10.15 (a) $\alpha = 0.0718$; (b) $\beta = 0.1151$

10.17 (a) $\alpha = 0.0384$; (b) $\beta = 0.5$; $\beta = 0.2776$

10.19 $z = -2.76$; yes, $\mu < 40$ months; P-value $= 0.0029$

10.21 $z = -1.64$; P-value $= 0.10$

10.23 $t = 0.77$; fail to reject H_0.

10.25 $z = 8.97$; yes, $\mu > 20,000$ kilometers; P-value < 0.001

10.27 $t = 12.72$; P-value < 0.0005; reject H_0.

10.29 $t = -1.98$; P-value $= 0.0312$; reject H_0

10.31 $z = -2.60$; conclude $\mu_A - \mu_B \leq 12$ kilograms.

10.33 $t = 1.50$; there is not sufficient evidence to conclude that the increase in substrate concentration would cause an increase in the mean velocity of more than 0.5 micromole per 30 minutes.

10.35 $t = 0.70$; there is not sufficient evidence to support the conclusion that the serum is effective.

10.37 $t = 2.55$; reject H_0: $\mu_1 - \mu_2 > 4$ kilometers.

10.39 $t' = 0.22$; fail to reject H_0.

10.41 $t' = 2.76$; reject H_0.

10.43 $t = -2.53$; reject H_0; the claim is valid.

10.45 $t = 2.48$; P-value < 0.02; reject H_0.

10.47 $n = 6$

10.49 $78.28 \approx 79$

10.51 5

10.53 (a) H_0: $M_{\text{hot}} - M_{\text{cold}} = 0$,
H_1: $M_{\text{hot}} - M_{\text{cold}} \neq 0$;
(b) paired t, $t = 0.99$; P-value > 0.30; fail to reject H_0.

10.55 P-value $= 0.4044$ (with a one-tailed test); the claim is not refuted.

10.57 $z = 1.44$; fail to reject H_0.

10.59 $z = -5.06$ with P-value ≈ 0; conclude that fewer than one-fifth of the homes are heated by oil.

10.61 $z = 0.93$ with P-value $= P(Z > 0.93) = 0.1762$; there is not sufficient evidence to conclude that the new medicine is effective.

10.63 $z = 2.36$ with P-value $= 0.0182$; yes, the difference is significant.

10.65 $z = 1.10$ with P-value $= 0.1357$; we do not have sufficient evidence to conclude that breast cancer is more prevalent in the urban community.

10.67 $\chi^2 = 18.13$ with P-value $= 0.0676$ (from computer output); do not reject H_0: $\sigma^2 = 0.03$.

10.69 $\chi^2 = 63.75$ with P-value $= 0.8998$ (from computer output); do not reject H_0.

10.71 $\chi^2 = 42.37$ with P-value $= 0.0117$ (from computer output); machine is out of control.

10.73 $f = 1.33$ with P-value $= 0.3095$ (from computer output); fail to reject H_0: $\sigma_1 = \sigma_2$.

10.75 $f = 0.086$ with P-value $= 0.0328$ (from computer output); reject H_0: $\sigma_1 = \sigma_2$ at level greater than 0.0328.

10.77 $f = 19.67$ with P-value $= 0.0008$ (from computer output); reject H_0: $\sigma_1 = \sigma_2$.

10.79 $\chi^2 = 10.14$; reject H_0, the ratio is not 5:2:2:1.

10.81 $\chi^2 = 4.47$; there is not sufficient evidence to claim that the die is unbalanced.

10.83 $\chi^2 = 3.125$; do not reject H_0: geometric distribution.

10.85 $\chi^2 = 5.19$; do not reject H_0: normal distribution.

10.87 $\chi^2 = 5.47$; do not reject H_0.

10.89 $\chi^2 = 124.59$; yes, occurrence of these types of crime is dependent on the city district.

10.91 $\chi^2 = 5.92$ with P-value $= 0.4332$; do not reject H_0.

10.93 $\chi^2 = 31.17$ with P-value < 0.0001; attitudes are not homogeneous.

10.95 $\chi^2 = 1.84$; do not reject H_0.

Chapter 11

11.1 (a) $b_0 = 64.529$, $b_1 = 0.561$;
(b) $\hat{y} = 81.4$

11.3 (a) $\hat{y} = 5.8254 + 0.5676x$;
(c) $\hat{y} = 34.205$ at 50°C

11.5 (a) $\hat{y} = 6.4136 + 1.8091x$;
(b) $\hat{y} = 9.580$ at temperature 1.75

11.7 (b) $\hat{y} = 31.709 + 0.353x$

11.9 (b) $\hat{y} = 343.706 + 3.221x$;
(c) $\hat{y} = \$456$ at advertising costs $= \$35$

11.11 (b) $\hat{y} = -1847.633 + 3.653x$

11.13 (a) $\hat{y} = 153.175 - 6.324x$;
(b) $\hat{y} = 123$ at $x = 4.8$ units

11.15 (a) $s^2 = 176.4$;
(b) $t = 2.04$; fail to reject H_0: $\beta_1 = 0$.

11.17 (a) $s^2 = 0.40$;
(b) $4.324 < \beta_0 < 8.503$;
(c) $0.446 < \beta_1 < 3.172$

11.19 (a) $s^2 = 6.626$;
(b) $2.684 < \beta_0 < 8.968$;
(c) $0.498 < \beta_1 < 0.637$

11.21 $t = -2.24$; reject H_0 and conclude $\beta < 6$

11.23 (a) $24.438 < \mu_{Y|24.5} < 27.106$;
(b) $21.88 < y_0 < 29.66$

11.25 $7.81 < \mu_{Y|1.6} < 10.81$

11.27 (a) 17.1812 mpg;
(b) no, the 95% confidence interval on mean mpg is $(27.95, 29.60)$;
(c) miles per gallon will likely exceed 18.

11.29 (b) $\hat{y} = 3.4156x$

11.31 The f-value for testing the lack of fit is 1.58, and the conclusion is that H_0 is not rejected. Hence, the lack-of-fit test is insignificant.

11.33 (a) $\hat{y} = 2.003x$;
(b) $t = 1.40$, fail to reject H_0.

11.35 $f = 1.71$ and P-value $= 0.2517$; the regression is linear.

11.37 (a) $b_0 = 10.812$, $b_1 = -0.3437$;
(b) $f = 0.43$; the regression is linear.

11.39 (a) $\hat{P} = -11.3251 - 0.0449T$;
(b) yes;
(c) $R^2 = 0.9355$;
(d) yes

11.41 (b) $\hat{N} = -175.9025 + 0.0902Y$; $R^2 = 0.3322$

11.43 $r = 0.240$

11.45 (a) $r = -0.979$;
(b) P-value $= 0.0530$; do not reject H_0 at 0.025 level;
(c) 95.8%

11.47 (a) $r = 0.784$;
(b) reject H_0 and conclude that $\rho > 0$;
(c) 61.5%.

Chapter 12

12.1 $\hat{y} = 0.5800 + 2.7122x_1 + 2.0497x_2$

12.3 (a) $\hat{y} = 27.547 + 0.922x_1 + 0.284x_2$;
(b) $\hat{y} = 84$ at $x_1 = 64$ and $x_2 = 4$

12.5 (a) $\hat{y} = -102.7132 + 0.6054x_1 + 8.9236x_2 + 1.4374x_3 + 0.0136x_4$;
(b) $\hat{y} = 287.6$

12.7 $\hat{y} = 141.6118 - 0.2819x + 0.0003x^2$

12.9 (a) $\hat{y} = 56.4633 + 0.1525x - 0.00008x^2$;
(b) $\hat{y} = 86.7\%$ when temperature is at 225°C

12.11 $\hat{y} = -6.5122 + 1.9994x_1 - 3.6751x_2 + 2.5245x_3 + 5.1581x_4 + 14.4012x_5$

12.13 (a) $\hat{y} = 350.9943 - 1.2720x_1 - 0.1539x_2$;
(b) $\hat{y} = 140.9$

12.15 $\hat{y} = 3.3205 + 0.4210x_1 - 0.2958x_2 + 0.0164x_3 + 0.1247x_4$

12.17 0.1651

12.19 242.72

12.21 (a) $\hat{\sigma}^2_{B_2} = 28.0955$; (b) $\hat{\sigma}_{B_1B_2} = -0.0096$

12.23 $t = 5.91$ with P-value $= 0.0002$. Reject H_0 and claim that $\beta_1 \neq 0$.

12.25 $0.4516 < \mu_{Y|x_1=900,x_2=1} < 1.2083$ and $-0.1640 < y_0 < 1.8239$

12.27 $263.7879 < \mu_{Y|x_1=75,x_2=24,x_3=90,x_4=98} < 311.3357$ and $243.7175 < y_0 < 331.4062$

12.29 (a) $t = -1.09$ with P-value $= 0.3562$;
(b) $t = -1.72$ with P-value $= 0.1841$;
(c) yes; not sufficient evidence to show that x_1 and x_2 are significant

12.31 $R^2 = 0.9997$

12.33 $f = 5.106$ with P-value $= 0.0303$; the regression is not significant at level 0.01.

12.35 $f = 34.90$ with P-value $= 0.0002$; reject H_0 and conclude $\beta_1 > 0$.

12.37 $f = 10.18$ with P-value < 0.01; x_1 and x_2 are significant in the presence of x_3 and x_4.

12.39 The two-variable model is better.

12.41 First model: $R^2_{\text{adj}} = 92.7\%$, C.V. $= 9.0385$.
Second model: $R^2_{\text{adj}} = 98.1\%$, C.V. $= 4.6287$.
The partial F-test shows P-value $= 0.0002$; model 2 is better.

12.43 Using x_2 alone is not much different from using x_1 and x_2 together since the R^2_{adj} are 0.7696 versus 0.7591, respectively.

12.45 (a) $\widehat{\text{mpg}} = 5.9593 - 0.00003773$ odometer $+ 0.3374$ octane $- 12.6266z_1 - 12.9846z_2$;
(b) sedan;
(c) they are not significantly different.

12.47 (b) $\hat{y} = 4.690$ seconds;
(c) $4.450 < \mu_{Y|\{180,260\}} < 4.930$

12.49 $\hat{y} = 2.1833 + 0.9576x_2 + 3.3253x_3$

12.51 (a) $\hat{y} = -587.211 + 428.433x$;
(b) $\hat{y} = 1180 - 191.691x + 35.20945x^2$;
(c) quadratic model

12.53 $\hat{\sigma}^2_{B_1} = 20{,}588$; $\hat{\sigma}^2_{B_{11}} = 62.6502$; $\hat{\sigma}_{B_1,B_{11}} = -1103.5$

12.55 (a) Intercept model is the best.

12.57 (a) $\hat{y} = 3.1368 + 0.6444x_1 - 0.0104x_2 + 0.5046x_3 - 0.1197x_4 - 2.4618x_5 + 1.5044x_6$;
(b) $\hat{y} = 4.6563 + 0.5133x_3 - 0.1242x_4$;
(c) C_p criterion: variables x_1 and x_2 with $s^2 = 0.7317$ and $R^2 = 0.6476$; s^2 criterion: variables x_1, x_3 and x_4 with $s^2 = 0.7251$ and $R^2 = 0.6726$;
(d) $\hat{y} = 4.6563 + 0.5133x_3 - 0.1242x_4$; This one does not lose much in s^2 and R^2;
(e) two observations have large R-Student values and should be checked.

12.59 (a) $\hat{y} = 125.8655 + 7.7586x_1 + 0.0943x_2 - 0.0092x_1x_2$;
(b) the model with x_2 alone is the best.

12.61 (a) $\hat{p} = (1 + e^{2.9949 - 0.0308x})^{-1}$;
(b) 1.8515

Chapter 13

13.1 $f = 0.31$; not sufficient evidence to support the hypothesis that there are differences among the 6 machines.

13.3 $f = 14.52$; yes, the difference is significant.

13.5 $f = 8.38$; the average specific activities differ significantly.

13.7 $f = 2.25$; not sufficient evidence to support the hypothesis that the different concentrations of $MgNH_4PO_4$ significantly affect the attained height of chrysanthemums.

13.9 $b = 0.79 > b_4(0.01, 4, 4, 4, 9) = 0.4939$. Do not reject H_0. There is not sufficent evidence to claim that variances are different.

13.11 $b = 0.7822 < b_4(0.05, 9, 8, 15) = 0.8055$. The variances are significantly different.

13.13 (a) P-value < 0.0001, significant,

(b) for contrast 1 vs. 2, P-value $< 0.0001Z$, significantly different; for contrast 3 vs. 4, P-value $= 0.0648$, not significantly different

13.15 Results of Tukey's tests are given below.

$\bar{y}_{4.}$	$\bar{y}_{3.}$	$\bar{y}_{1.}$	$\bar{y}_{5.}$	$\bar{y}_{2.}$
2.98	4.30	5.44	6.96	7.90

13.17 (a) P-value $= 0.0121$; yes, there is a significant difference.

(b)

Depletion	Modified Hess	Substrate Removal Kicknet	Surber	Kicknet

13.19 $f = 70.27$ with P-value < 0.0001; reject H_0.

$\bar{x}_0$	$\bar{x}_{25}$	$\bar{x}_{100}$	$\bar{x}_{75}$	$\bar{x}_{50}$
55.167	60.167	64.167	70.500	72.833

Temperature is important; both 75° and 50°(C) yielded batteries with significantly longer activated life.

13.21 The mean absorption is significantly lower for aggregate 4 than for the other aggregates.

13.23 Comparing the control to 1 and 2: significant; comparing the control to 3 and 4: insignificant

13.25 $f(\text{fertilizer}) = 6.11$; there is significant difference among the fertilizers.

13.27 $f = 5.99$; percent of foreign additives is not the same for all three brands of jam; brand A

13.29 P-value < 0.0001; significant

13.31 P-value $= 0.0023$; significant

13.33 P-value $= 0.1250$; not significant

13.35 P-value < 0.0001;
$f = 122.37$; the amount of dye has an effect on the color of the fabric.

13.37 (a) $y_{ij} = \mu + A_i + \epsilon_{ij}$, $A_i \sim n(x; 0, \sigma_\alpha)$, $\epsilon_{ij} \sim n(x; 0, \sigma)$;

(b) $\hat{\sigma}^2_\alpha = 0$ (the estimated variance component is -0.00027); $\hat{\sigma}^2 = 0.0206$.

13.39 (a) $f = 14.9$; operators differ significantly;

(b) $\hat{\sigma}^2_\alpha = 28.91$; $s^2 = 8.32$.

13.41 (a) $y_{ij} = \mu + A_i + \epsilon_{ij}$, $A_i \sim n(x; 0, \sigma_\alpha)$;

(b) yes; $f = 5.63$ with P-value $= 0.0121$;

(c) there is a significant loom variance component.

Chapter 14

14.1 (a) $f = 8.13$; significant;

(b) $f = 5.18$; significant;

(c) $f = 1.63$; insignificant

14.3 (a) $f = 14.81$; significant;

(b) $f = 9.04$; significant;

(c) $f = 0.61$; insignificant

14.5 (a) $f = 34.40$; significant;

(b) $f = 26.95$; significant;

(c) $f = 20.30$; significant

14.7 Test for effect of temperature: $f_1 = 10.85$ with P-value $= 0.0002$;
Test for effect of amount of catalyst: $f_2 = 46.63$ with P-value < 0.0001;
Test for effect of interaction: $f = 2.06$ with P-value $= 0.074$.

14.9 (a)

Source of Variation	**df**	**Sum of Squares**	**Mean Squares**	f	P
Cutting speed	1	12.000	12.000	1.32	0.2836
Tool geometry	1	675.000	675.000	74.31	< 0.0001
Interaction	1	192.000	192.000	21.14	0.0018
Error	8	72.667	9.083		
Total	11	951.667			

(b) The interaction effect masks the effect of cutting speed;

(c) $f_{\text{tool geometry}=1} = 16.51$ and P-value $= 0.0036$;
$f_{\text{tool geometry}=2} = 5.94$ and P-value $= 0.0407$.

14.11 (a)

Source of Variation	df	Sum of Squares	Mean Squares	f	P
Method	1	0.000104	0.000104	6.57	0.0226
Laboratory	6	0.008058	0.001343	84.70	< 0.0001
Interaction	6	0.000198	0.000033	2.08	0.1215
Error	14	0.000222	0.000016		
Total	27	0.008582			

(b) The interaction is not significant;

(c) Both main effects are significant;

(e) $f_{\text{laboratory}=1} = 0.01576$ and P-value $= 0.9019$; no significant difference between the methods in laboratory 1;
$f_{\text{tool geometry}=2} = 9.081$ and P-value $= 0.0093$.

14.13 (b)

Source of Variation	df	Sum of Squares	Mean Squares	f	P
Time	1	0.060208	0.060208	157.07	< 0.0001
Treatment	1	0.060208	0.060208	157.07	< 0.0001
Interaction	1	0.000008	0.000008	.02	0.8864
Error	8	0.003067	0.000383		
Total	11	0.123492			

(c) Both time and treatment influence the magnesium uptake significantly, although there is no significant interaction between them.

(d) $Y = \mu + \beta_T \text{Time} + \beta_Z Z + \beta_{TZ} \text{Time } Z + \epsilon$, where $Z = 1$ when treatment $= 1$ and $Z = 0$ when treatment $= 2$;

(e) $f = 0.02$ with P-value $= 0.8864$; the interaction in the model is insignificant.

14.15 (a) Interaction is significant at a level of 0.05, with P-value of 0.0166.

(b) Both main effects are significant.

14.17 (a) AB: $f = 3.83$; significant;
AC: $f = 3.79$; significant;
BC: $f = 1.31$; not significant;
ABC: $f = 1.63$; not significant;

(b) A: $f = 0.54$; not significant;
B: $f = 6.85$; significant;
C: $f = 2.15$; not significant;

(c) The presence of AC interaction masks the main effect C.

14.19 (a) Stress: $f = 45.96$ with P-value < 0.0001;
coating: $f = 0.05$ with P-value $= 0.8299$;
humidity: $f = 2.13$ with P-value $= 0.1257$;
coating × humidity: $f = 3.41$ with P-value $= 0.0385$;
coating × stress: $f = 0.08$ with P-value $= 0.9277$;
humidity × stress: $f = 3.15$ with P-value $= 0.0192$;
coating × humidity × stress: $f = 1.93$ with P-value $= 0.1138$.

(b) The best combination appears to be uncoated, medium humidity, and a stress level of 20.

14.21

Effect	f	P
Temperature	14.22	< 0.0001
Surface	6.70	0.0020
HRC	1.67	0.1954
T × S	5.50	0.0006
T × HRC	2.69	0.0369
S × HRC	5.41	0.0007
T × S × HRC	3.02	0.0051

14.23 (a) Yes; brand × type; brand × temperature;

(b) yes;

(c) brand Y, powdered detergent, hot temperature.

14.25 (a)

Effect	f	P
Time	543.53	< 0.0001
Temp	209.79	< 0.0001
Solvent	4.97	0.0457
Time × Temp	2.66	0.1103
Time × Solvent	2.04	0.1723
Temp × Solvent	0.03	0.8558
Time × Temp × Solvent	6.22	0.0140

Although three two-way interactions are shown to be insignificant, they may be masked by the significant three-way interaction.

14.27 (a) $f = 1.49$; no significant interaction;

(b) $f(\text{operators}) = 12.45$; significant;
$f(\text{filters}) = 8.39$; significant;

(c) $\hat{\sigma}^2_\alpha = 0.1777$ (filters);
$\hat{\sigma}^2_\beta = 0.3516$ (operators);
$s^2 = 0.185$

14.29 (a) $\hat{\sigma}^2_\beta$, $\hat{\sigma}^2_\gamma$, $\hat{\sigma}^2_{\alpha\gamma}$ are significant;

(b) $\hat{\sigma}^2_\gamma$ and $\hat{\sigma}^2_{\alpha\gamma}$ are significant.

14.31 (a) Mixed model;

(b) Material: $f = 47.42$ with P-value < 0.0001;
brand: $f = 1.73$ with P-value $= 0.2875$;
material $\times$ brand: $f = 16.06$ with P-value $= 0.0004$;

(c) no

Chapter 15

15.1 B and C are significant at level 0.05.

15.3 Factors A, B, and C have negative effects on the phosphorus compound, and factor D has a positive effect. However, the interpretation of the effect of individual factors should involve the use of interaction plots.

15.5 Significant effects:
A: $f = 9.98$; BC: $f = 19.03$.
Insignificant effects:
B: $f = 0.20$; C: $f = 6.54$; D: $f = 0.02$; AB: $f = 1.83$;
AC: $f = 0.20$; AD: $f = 0.57$; BD: $f = 1.83$;
CD: $f = 0.02$. Since the BC interaction is significant, both B and C would be investigated further.

15.9 (a) $b_A = 5.5$, $b_B = -3.25$, and $b_{AB} = 2.5$;

(b) the values of the coefficients are one-half those of the effects;

(c) $t_A = 5.99$ with P-value $= 0.0039$;
$t_B = -3.54$ with P-value $= 0.0241$;
$t_{AB} = 2.72$ with P-value $= 0.0529$;
$t^2 = F$.

15.11 (a) $A = -0.8750$, $B = 5.8750$, $C = 9.6250$, $AB = -3.3750$, $AC = -9.6250$, $BC = 0.1250$, and $ABC = -1.1250$;
B, C, AB, and AC appear important based on their magnitude.

(b)

Effects	***P*-Value**
A	0.7528
B	0.0600
C	0.0071
AB	0.2440
AC	0.0071
BC	0.9640
ABC	0.6861

(c) Yes;

(d) At a high level of A, C essentially has no effect. At a low level of A, C has a positive effect.

15.13 (a)

Machine			
1	**2**	**3**	**4**
(1)	c	a	ac
ab	d	b	ad
cd	e	acd	ae
ce	abc	ace	bc
de	abd	ade	bd
$abcd$	abe	bcd	be
$abce$	cde	bce	$acde$
$abde$	$abcde$	bde	$bcde$

(b) ABD, CDE, $ABCDE$ (one possible design)

15.15 (a) x_2, x_3, x_1x_2, and x_1x_3;

(b) Curvature: P-value $= 0.0038$;

(c) One additional design point different from the original ones

15.17 $(0, -1)$, $(0, 1)$, $(-1, 0)$, $(1, 0)$ might be used.

15.19 (a) With BCD as the defining contrast, the principal block contains (1), a, bc, abc, bd, abd, cd, acd;

(b)

Block 1	Block 2
(1)	a
bc	abc
abd	bd
acd	cd

confounded by ABC;

(c) Defining contrast BCD produces the following aliases: $A \equiv ABCD$, $B \equiv CD$, $C \equiv BD$, $D \equiv BC$, $AB \equiv ACD$, $AC \equiv ABD$, and $AD \equiv ABC$. Since AD and ABC are confounded with blocks, there are only 2 degrees of freedom for error from the interactions not confounded.

Source of Variation	**Degrees of Freedom**
A	1
B	1
C	1
D	1
Blocks	1
Error	2
Total	7

15.21 (a) With the defining contrasts $ABCE$ and $ABDF$, the principal block contains (1), ab, acd, bcd, ce, $abce$, ade, bde, acf, bcf, df, $abdf$, aef, bef, $cdef$, $abcdef$;

(b) $A \equiv BCE \equiv BDF \equiv ACDEF$,
$AD \equiv BCDE \equiv BF \equiv ACEF$,
$B \equiv ACE \equiv ADF \equiv BCDEF$,
$AE \equiv BC \equiv BDEF \equiv ACDF$,
$C \equiv ABE \equiv ABCDF \equiv DEF$,
$AF \equiv BCEF \equiv BD \equiv ACDE$,
$D \equiv ABCDE \equiv ABF \equiv CEF$,
$CE \equiv AB \equiv ABCDEF \equiv DF$,
$E \equiv ABC \equiv ABDEF \equiv CDF$,
$DE \equiv ABCD \equiv ABEF \equiv CF$,
$F \equiv ABCEF \equiv ABD \equiv CDE$,
$BCD \equiv ADE \equiv ACF \equiv BEF$,
$AB \equiv CE \equiv DF \equiv ABCDEF$,
$BCF \equiv AEF \equiv ACD \equiv BDE$,
$AC \equiv BE \equiv BCDF \equiv ADEF$;

Source of Variation	Degrees of Freedom
A	1
B	1
C	1
D	1
E	1
F	1
AB	1
AC	1
AD	1
BC	1
BD	1
CD	1
CF	1
Error	2
Total	15

15.23

Source	df	SS	MS	f	P
A	1	6.1250	6.1250	5.81	0.0949
B	1	0.6050	0.6050	0.57	0.5036
C	1	4.8050	4.8050	4.56	0.1223
D	1	0.2450	0.2450	0.23	0.6626
Error	3	3.1600	1.0533		
Total	7	14.9400			

15.25

Source	df	SS	MS	f	P
A	1	388,129.00	388,129.00	3585.49	0.0001
B	1	277,202.25	277,202.25	2560.76	0.0001
C	1	4692.25	4692.25	43.35	0.0006
D	1	9702.25	9702.25	89.63	0.0001
E	1	1806.25	1806.25	16.69	0.0065
AD	1	1406.25	1406.25	12.99	0.0113
AE	1	462.25	462.25	4.27	0.0843
BD	1	1156.00	1156.00	10.68	0.0171
BE	1	961.00	961.00	8.88	0.0247
Error	6	649.50	108.25		
Total	15	686,167.00			

All main effects are significant at the 0.05 level; AD, BD, and BE are also significant at the 0.05 level.

15.27 The principal block contains af, be, cd, abd, ace, bcf, def, $abcdef$.

15.29 $A \equiv BD \equiv CE \equiv CDF \equiv BEF \equiv ABCF \equiv ADEF \equiv ABCDE$;
$B \equiv AD \equiv CF \equiv CDE \equiv AEF \equiv ABCE \equiv BDEF \equiv ABCDF$;
$C \equiv AE \equiv BF \equiv BDE \equiv ADF \equiv CDEF \equiv ABCD \equiv ABCEF$;
$D \equiv AB \equiv EF \equiv BCE \equiv ACF \equiv BCDF \equiv ACDE \equiv ABDEF$;
$E \equiv AC \equiv DF \equiv ABF \equiv BCD \equiv ABDE \equiv BCEF \equiv ACDEF$;
$F \equiv BC \equiv DE \equiv ACD \equiv ABE \equiv ACEF \equiv ABDF \equiv BCDEF$.

15.31 $x_1 = 1$ and $x_2 = 1$

15.33 (a) Yes;
(b) (i) $E(\hat{y}) = 79.00 + 5.281A$;
(ii) $\text{Var}(\hat{y}) = 6.22^2\sigma_Z^2 + 5.70^2A^2\sigma_Z^2 + 2(6.22)(5.70)A\sigma_Z^2$;
(c) velocity at low level;
(d) velocity at low level;
(e) yes

15.35 $\hat{y} = 12.7519 + 4.7194x_1 + 0.8656x_2 - 1.4156x_3$; units are centered and scaled; test for lack of fit, $F = 81.58$, with P-value < 0.0001.

15.37 AFG, BEG, CDG, DEF, $CEFG$, $BDFG$, $BCDE$, $ADEG$, $ACDF$, $ABEF$, and $ABCDEFG$

Chapter 16

16.1 $x = 7$ with P-value $= 0.1719$; fail to reject H_0.

16.3 $x = 3$ with P-value $= 0.0244$; reject H_0.

16.5 $x = 4$ with P-value $= 0.3770$; fail to reject H_0.

16.7 $x = 4$ with P-value $= 0.1335$; fail to reject H_0.

16.9 $w = 43$; fail to reject H_0.

16.11 $w_+ = 17.5$; fail to reject H_0.

16.13 $w_+ = 15$ with $n = 13$; reject H_0 in favor of $\tilde{\mu}_1 - \tilde{\mu}_2 < 8$.

16.15 $u_1 = 4$; claim is not valid.

16.17 $u_2 = 5$; A operates longer.

16.19 $u = 15$; fail to reject H_0.

16.21 $h = 10.58$; operating times are different.

16.23 $v = 7$ with P-value $= 0.910$; random sample.

16.25 $v = 6$ with P-value $= 0.044$; fail to reject H_0.

16.27 $v = 4$; random sample.

16.29 0.70

16.31 0.995

16.33 (a) $r_s = 0.39$; (b) fail to reject H_0.

16.35 (a) $r_s = 0.72$; (b) reject H_0, so $\rho > 0$.

16.37 (a) $r_s = 0.71$; (b) reject H_0, so $\rho > 0$.

Chapter 18

18.1 $p^* = 0.173$

18.3 (a) $\pi(p \mid x = 1) = 40p(1-p)^3/0.2844$; $0.05 < p < 0.15$;
(b) $p^* = 0.106$

18.5 (a) $beta(95, 45)$; (b) 1

18.7 $8.077 < \mu < 8.692$

18.9 (a) 0.2509; (b) $68.71 < \mu < 71.69$; (c) 0.0174

18.13 $p^* = \frac{6}{x+2}$

18.15 2.21

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